

AR Project Report

Realise your dreams & become a Painter



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Application Overview

Our project allows the user to become a painter without the fear of making mistakes or damaging their real environment with paint.

This application is an Augmented Reality project developed for the Meta Quest 3 headset. When launching the application, passthrough will allow the user to see the real environment and canvas will appear. You then will be able to paint on it.

The user can interact with several objects and modify different settings through direct hand interactions with virtual elements.

Technical details

The project has been developed using Unity 2022.3.62f2 and the MetaXR All-in-One SDK.

Here is a link to the github repository: [Astri2/FIB-VAR-AR-Project](#). You can clone it.

Here is a link to the apk download: [Releases · Astri2/FIB-VAR-AR-Project](#). It has been tested on the Meta Quest 3 headset.

Virtual objects of the application

When using the application, the user will see many virtual objects appearing:

- A brush
- A colour picker
- A rubber
- A canvas
- A button on a pedestal
- Some cookies

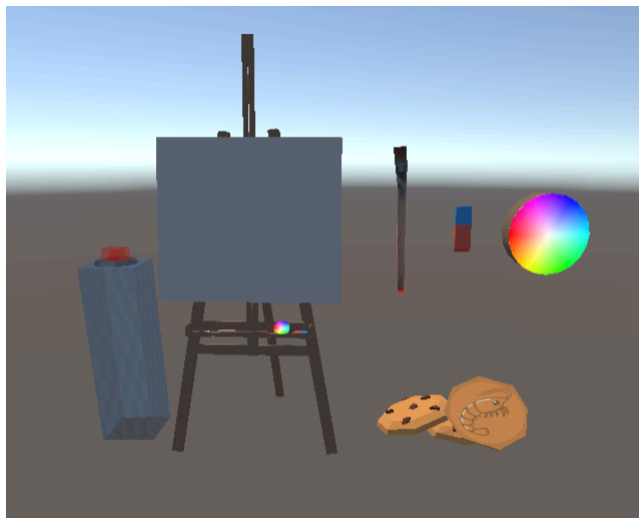


Figure 1 - Virtual objects of the application

Tutorial

First you have to make sure that the room has been scanned through the Meta Quest 3 environment setup.

In this application you can grab several virtual elements without the use of the controllers. It uses hand detection, allowing you to interact with elements as you would do with real objects. You will be able to grab a brush, a colour picker and a rubber.

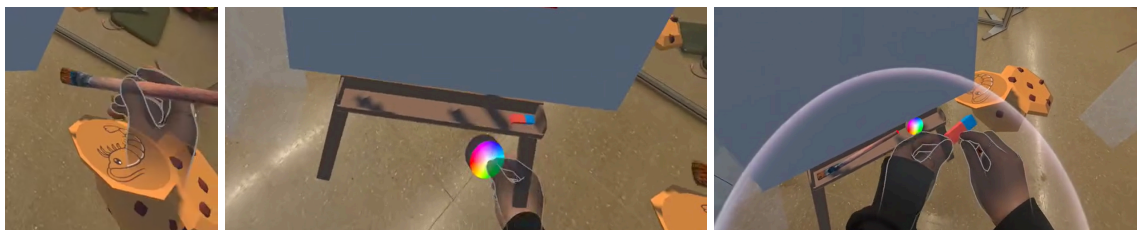


Figure 2 - Grabbing virtual objects

The brush is the tool that will give you the possibility of painting on almost everything.

It is possible for you to paint on virtual elements. By approaching your brush of canvas, pedestal and cookies, colours will appear where you touch them.



Figure 3 - Painting on virtual objects

But it is also possible to paint the real world! Indeed, paint will also appear if you approach your brush of the floor, the walls, the ceiling, tables or any object that has been detected inside your room.

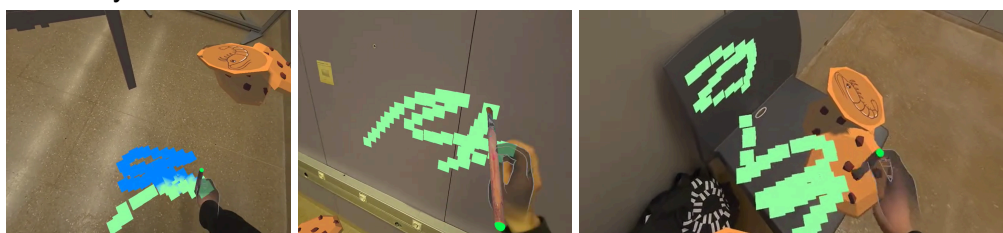


Figure 4 - Painting on the real world

The colour picker and the rubber will allow you to customize the aspect of your painting. By approaching the brush of the colour picker, you will be able to choose the colour of your paint on the brush. It will not change the colour of the paint you have already applied.



Figure 5 - Changing the colour of the paint

If you made a mistake, it is also possible for you to approach the brush from the rubber to enable erase mode. It will then be possible to erase some parts of your painting (both on real objects and canvas) with the brush (you can go back to paint mode by selecting a color). In order to erase the paint on the entire canvas, you can press the button you see on its left.



Figure 6 - Erase mode



Figure 7 - Clearing the entire canvas

The colour pickers and the rubbers appear under the canvas next to the brush. They are small in order to fit but you can rescale them (not the brush) as you want by grabbing them with both hands. The size of the canvas can also be modified the same way.



Figure 8 - Rescaling of the colour picker



Figure 9 - Rescaling of the rubber

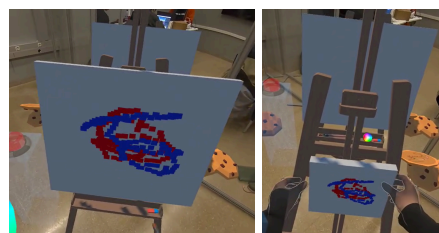


Figure 10 - Rescaling of the canvas

When you move in your physical room, you may exit the white circle which represents your game zone. That is not a problem, simply go where you wanted to go, then click on create a new boundary

Interactions

As stated before there are 3 types of interactions in this application.

1. Interaction of the user with virtual elements through real hand recognition
 - Grabbing virtual elements (brush, colour picker, rubber)
 - Pushing a button
 - Rescaling virtual elements (canvas, colour picker, rubber)
2. Interaction between virtual elements
 - Choosing the colour of the paint
 - Painting on virtual elements (canvas, cookies, button pedestals)
3. Interaction between virtual and real elements
 - Painting the real environment (tables, walls, floor, ceiling, ...)

Real Scene Recognition

For this project we used passthrough in order to visualize virtual elements in the real world. After scanning the room with the Meta Quest 3 headset, we used the real scene recognition system to place our virtual objects (the canvas can only be placed on the floor, while the other objects can be placed on tables). We added colliders to the planes and volumes created by the system, which allowed us to create the interaction of painting on them with the virtual brush.

Control Mechanisms

As shown above, several control mechanisms have been implemented:

- Changing the colour of the paint.
- Switching between paint and erase mode.
- Clearing canvas.

Demo

Here is a link to the demo video: https://www.youtube.com/watch?v=tCdbere_eMw